NIT Admin & API - Hosting Guide

Nesma Infrastructure & Technology · Production server runbook · No passwords in this document

1. Overview

Item	Value
Admin panel	`https://nit-admin.zakhf.com/`
Website API	`https://nit-admin.zakhf.com/api/`
Frontend (Next.js)	`https://nit-hazel.vercel.app`
Server	Hostinger VPS, `88.222.214.123`, Ubuntu 24.04 LTS (2 CPU, 8 GB RAM)
Stack	Django 4.2 · Python 3.12 · PostgreSQL 16 · Redis · Gunicorn · Nginx · Let's Encrypt
Linux user	`nesma` (SSH key only, no password)
Repository	`https://github.com/jaseemrahman/NESMA-BACKEND` (branch `main`)
Deploys	Automatic on every push to `main` (GitHub Actions), after tests pass

The server also hosts other projects (BlueDot, NKE, Zuarak, Eclat, Dojo). NIT is fully isolated: own Linux user, folder, database, Redis database, service and Nginx site. Nothing of the other projects was changed.

2. Server access

Purpose	Command / location
Administration (root)	`ssh root@88.222.214.123` (SSH key)
Run a command as the app user	`sudo -u nesma <command>` (from root)
CI deploy key	`~/.ssh/nesma_deploy` on the maintainer's Mac (GitHub secret `SSH_KEY`)

Where the secrets are:

- Django `SECRET_KEY`, database password, SMTP password: only in `/var/www/nesma/app/.env` (mode 600, owner `nesma`).
- GitHub: repository secrets `HOST`, `USERNAME`, `SSH_KEY`.
- Never commit `.env`; it is in `.gitignore`.

The `nesma` user's SSH key is restricted with a forced command: it can only run the deploy script (no shell, no port forwarding). To revoke CI access, delete its line in `/home/nesma/.ssh/authorized_keys`.

3. Folder structure (server)

```
/var/www/nesma/                      owner nesma, group www-data
├── app/                             git checkout of the repository (branch main)
│   ├── .env                         production settings + secrets (chmod 600)
│   ├── config/                      Django settings, urls, wsgi
│   ├── superadmin/                  admin panel app (models, views, PDF export, migrations)
│   ├── website/                     public REST API app
│   ├── templates/                   admin + email templates
│   ├── static/                      static source files (collected into ../static)
│   ├── deploy/                      deploy.sh, nesma.service, nginx + sudoers templates, this guide
│   └── manage.py
├── venv/                            Python virtual environment
├── static/                          collectstatic output - served by Nginx (cached 1 year)
├── media/                           uploads: blogs/ public, careers/ CVs admin-only
└── cache/                           spare (cache uses Redis)
```

Other files created for NIT (and nothing else):

Path	What
`/etc/systemd/system/nesma.service`	Gunicorn service
`/run/nesma/gunicorn.sock`	Unix socket between Nginx and Gunicorn (no TCP port)
`/etc/nginx/sites-available/nesma` (+ link in `sites-enabled/`)	Nginx site incl. SSL (certbot)
`/etc/letsencrypt/live/nit-admin.zakhf.com/`	SSL certificate
`/etc/sudoers.d/nesma`	`nesma` may only restart / check its own service
`/etc/cron.daily/nesma-backup`	daily backup job
`/var/backups/nesma/`	backups (14 days)
`/home/nesma/.ssh/authorized_keys`	restricted CI deploy key
PostgreSQL `nesma_db` / role `nesma_user`	database
Redis database `7`, key prefix `nesma:`	cache (db 2 belongs to another project)

4. Application service (Gunicorn)

Task	Command (as root)
Status	`systemctl status nesma`
Restart (after .env change)	`systemctl restart nesma`
Live logs	`journalctl -u nesma -f`
Last 100 log lines	`journalctl -u nesma -n 100 --no-pager`
Enable at boot	`systemctl enable nesma` (already enabled)

Gunicorn runs 3 workers x 2 threads, 60 s timeout, workers recycled every ~1000 requests.

5. Nginx & SSL

Task	Command (as root)
Test config (always before reload)	`nginx -t`
Reload	`systemctl reload nginx`
Access / error logs	`/var/log/nginx/nesma.access.log` · `/var/log/nginx/nesma.error.log`
Certificate status	`certbot certificates --cert-name nit-admin.zakhf.com`
Test renewal	`certbot renew --dry-run --cert-name nit-admin.zakhf.com`

The certificate renews automatically (certbot timer). HTTP redirects to HTTPS; HSTS is enabled.

Upload limit: 15 MB (`client_max_body_size`).

6. Database (PostgreSQL 16)

Task	Command (as root)
Open psql	`sudo -u postgres psql nesma_db`
Size	`sudo -u postgres psql -c "SELECT pg_size_pretty(pg_database_size('nesma_db'))"`
Manual dump	`sudo -u postgres pg_dump --no-owner nesma_db | gzip > /root/nesma_$(date +%F).sql.gz`
Restore a dump	`gunzip -c FILE.sql.gz | sudo -u postgres psql nesma_db`

Only reachable on localhost. Connection string: `DATABASE_URL` in `.env`.

7. Django management commands

Run from root (the settings are read from `/var/www/nesma/app/.env`):

```
DJ="sudo -u nesma /var/www/nesma/venv/bin/python /var/www/nesma/app/manage.py"
$DJ createsuperuser                # email is required - the admin signs in with email
$DJ changepassword USERNAME
$DJ migrate
$DJ collectstatic --noinput
$DJ check --deploy
$DJ seed_demo_data --blogs-only    # the 6 posts from the website's media page (skips existing)
$DJ shell
```

Clear only NIT's cache (Redis db 7): `redis-cli -n 7 FLUSHDB`

8. Environment variables (.env)

Variable	Purpose
`SECRET_KEY`	Django secret (generated on the server)
`DEBUG`	`False` in production
`ALLOWED_HOSTS` / `CSRF_TRUSTED_ORIGINS`	`nit-admin.zakhf.com`
`DATABASE_URL`	PostgreSQL connection
`CACHE_URL` / `CACHE_KEY_PREFIX`	Redis db 7 / `nesma`
`CORS_ALLOW_ALL_ORIGINS`	`True` = any website may call the API
`SITE_URL` / `MEDIA_DOMAIN`	used for full https image URLs and email links
`STATIC_ROOT` / `MEDIA_ROOT`	`/var/www/nesma/static` / `/var/www/nesma/media`
`EMAIL_*`	SMTP for enquiry / career notifications
`ADMIN_NOTIFICATION_EMAIL`	receives enquiries and applications
`SUBMISSION_THROTTLE_RATE`	form limit per IP (default `10/hour`)
`CV_MAX_SIZE_MB` / `PDF_EXPORT_MAX_ROWS`	upload and export limits
`SECURE_SSL_REDIRECT` / `SECURE_HSTS_SECONDS`	HTTPS enforcement

After editing `.env`: `systemctl restart nesma`.

9. CI/CD - automatic deploys

Workflow: `.github/workflows/ci-cd.yml`

1. Every push and pull request to `main`: Django checks, migration check and the full test suite on PostgreSQL 16.

2. Push to `main` with green tests: GitHub validates the deploy key, connects to the server as `nesma` (server host key pinned) and runs `deploy/deploy.sh`.

3. `deploy.sh`: pull `main` → `pip install` → `check --deploy` → `migrate` → `collectstatic` → restart → health check on `/api/career-options/`. If the service is not healthy, the previous commit is restored automatically (database migrations are not reversed - keep migrations backward compatible).

GitHub secret	Value
`HOST`	`88.222.214.123`
`USERNAME`	`nesma`
`SSH_KEY`	private key `~/.ssh/nesma_deploy`

Manual deploy (same as CI): `sudo -u nesma bash /var/www/nesma/app/deploy/deploy.sh`

Roll back to an older commit by hand:

```
sudo -u nesma git -C /var/www/nesma/app reset --hard <commit>
systemctl restart nesma
```

10. Backups

`/etc/cron.daily/nesma-backup` runs every night and keeps 14 days in `/var/backups/nesma/`:

`nesma_db_YYYY-MM-DD.sql.gz` (database) and `nesma_media_YYYY-MM-DD.tar.gz` (uploads).

Run now: `/etc/cron.daily/nesma-backup`. Copy important backups off the server regularly.

11. Troubleshooting

Symptom	Check
502 Bad Gateway	`systemctl status nesma` and `journalctl -u nesma -n 100 --no-pager`
500 error	`journalctl -u nesma -n 100 --no-pager` (Django tracebacks are logged there)
Old CSS/JS after deploy	run `collectstatic`; file names are hashed so browsers refresh automatically
Images 404	file exists in `/var/www/nesma/media/`? Nginx `alias` path in `/etc/nginx/sites-available/nesma`
Emails not sent	`EMAIL_*` in `.env`; logs show `Background task ... failed`
GitHub deploy failed	Actions log → "Validate SSH key" / "Deploy over SSH" step; server: `journalctl -u ssh | grep nesma`
Disk / memory	`df -h` · `free -h` · `du -sh /var/www/nesma/* /var/backups/nesma`

12. Notes

- The repository's `deploy/nginx-nit-admin.zakhf.com.conf` is the pre-SSL template; the live file additionally contains the HTTPS block added by certbot.
- Other projects on this server were not modified. Observed during setup: the certificate for `bluedot.zuarak.com` had expired (30 Sep 2026) - unrelated to NIT.
- Rotate any passwords that were shared in chat or documents.